Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics

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Abstract

Observer Patch Holography (OPH) proposes that physics is what bounded observers can establish together: finite patches compare records on shared boundaries, repair disagreements, and treat as objective whatever survives every comparison. The universe, on this view, is a timeless self-referential structure whose observers recover its specification from their records and realize it, so that it reads itself into existence. Nothing lies outside to impose premises, so the recovered specification owes no a-priori justification; it has to run and reproduce what observers measure.

That specification is a candidate, an executable observer: a twelve-port carrier, a native repair law, retained records and record-controlled feedback, joined in one interacting quantum process of electric flux, charged matter and a scalar field whose feedback response is certified nonzero in the infinite-occupation limit. Finite theorems turn the specification's readouts into familiar structure: schedule-independent public records, the Born rule and Tsirelson bound on completed records, an intrinsic three-dimensional carrier, the Standard Model gauge Lie algebra with an anomaly-free generation, a source-derived $(1 + 3)$ causal precursor with proper-time ratios from record counts, and the Einstein equation from a realized null balance. We state the acceptance criteria of the construction, which premises it discharges, which it takes as declared inputs, and the frozen tests that can reject its branches.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock remains a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Time conventions. Host runtime, solver counters, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks remain declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

Keywords: foundations of physics; Observer Patch Holography; observer consensus; executable observer; self-reference; quantum records; gauge structure; Einstein field equation.

1 Introduction

Physics usually begins with a world and asks how observers fit into it. Observer Patch Holography (OPH) begins with observers and asks what world they can establish together. Its observers are finite. Each holds a patch of local state, reads a boundary it shares with its neighbors, keeps durable records, and repairs its state when records disagree; it accepts only repairs that preserve protected data. OPH identifies an objective fact with a record that survives every authorized comparison. Quantum probability, Lorentzian kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [17, 18, 19, 53, 60]; OPH asks whether they are different public readouts of one finite observer architecture. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret; no observer-independent continuum is assumed beneath them.

The proposal about the whole is closure. The universe is a timeless, self-referential structure whose observers recover the specification of their own world from completed records and build hardware whose execution realizes that same history, including its builders. The structure reads itself into existence. This changes what a premise has to do. A theory set against an external world must justify its postulates in advance or by fit; a structure with no outside has no place from which a postulate could be imposed. Whatever observers recover and implement therefore needs no further justification: the carrier, the observer algebra, the repair law and two closure coordinates are justified by being recovered, by running, and by reproducing what observers measure. The same rule cuts the other way. Nothing may be assumed that observers have not recovered, and every change of description they can make must be implemented by the structure's own operations. Agreement fixes what counts as shared, so "our universe" is the world described by the public record. The executable specification is where this account enters the loop; the loop itself has no first step. Section 6 develops the hypothesis and its consequences.

The programme's target follows from this reading. An executable construction should contain observers who make the measurements we make and who can use the effective continuum descriptions we use: quantum field theory, classical mechanics and dynamical gravity, each in its regime. The specification exists and runs. Its finite observer executes repair, writes and retains records and

chooses its next interaction from those records; an electromagnetic detector evaluates the coupling from a closure coordinate and feeds a single bit back; and one interacting quantum process joins electric flux, charged matter and a scalar field with complete measurements and a feedback response certified nonzero in the infinite-occupation limit. Section 4 turns six readouts of the specification into familiar structure: schedule-independent public records, the quantum identities of completed records, three spatial directions, the gauge algebra with one anomaly-free generation, a causal precursor with record-count clocks, and the Einstein equation up to a cosmological constant.

The results have different logical types, and Table 1 gives the classes used throughout; each result below carries one of them. Each step is identified by its artifact as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Table 1: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

Throughout, "source" means computed from the declared finite source specification, including every explicitly named source law, with no measured physical input. A physical realization map is not source-only merely because the finite branch calculation consumes no measured target value.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

How this paper is organized. This paper gives the account and develops selected results in detail. Section 2 describes the observer, its carrier and repair law, and what runs. Section 3 shows what observers agree on: public normal forms and the quantum identities of completed records. Section 4 develops five results in detail: the three spatial directions, the gauge algebra and one generation, measurement inside one interacting process, the source-derived causal precursor, and the Einstein equation; it lists the other results briefly. Section 5 states the completion target, the

acceptance criteria and their status. Section 6 develops closure. Section 7 gives the frozen tests, Section 8 the verification stack, and Section 9 the comparison with neighboring approaches. A companion construction paper states the executable specification and its certified components in full [66], and the theorem library carries the long proofs [1, 4, 5, 6, 8, 11, 9, 7, 2].

Boundary 1.2. This paper claims no completed derivation of particle masses, no source-only determination of the fine-structure constant, no completed cosmology, and no claim that formal verification establishes physical truth. Quantitative comparisons that consume measured values are labeled diagnostics. Section 7 states the frozen tests and the physical premises that set the scope of any verdict.

2 The observer

2.1 Patches, records and repair

Definition 2.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 2.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

2.2 The twelve-port carrier

Each local carrier has twelve primitive boundary ports with oriented icosahedral incidence: thirty seams and twenty faces, five seams at each port, every seam on two oppositely oriented faces. Carriers federate through typed seam algebras with coherent triple overlaps into a nerve carrying a degree-one bridge to an oriented spherical support. The theorem library calls this carrier, federation and response layer A1. Within the closure hypothesis of Section 6 it is the part of the specification that observers recover from their records; every result below displays its dependence on it.

Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{1}$$

On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real representations of real type; the projectors, characters and irreducibility certificates are machine-checked [48, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates.

2.3 The native repair law

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$, the native repair law, called C, moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}[d - (2c-1)(|d| \bmod 2)], \qquad y = x + j\,(e_b - e_a), \tag{2}$$

so load is conserved and the signed current j is the native observation. Its quantum current instrument has Kraus operators $L_{(c,j)} = 2^{-1/2} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|$. A second law, E, is C with one additional seam-parity record. Both laws have the same diagonal classical repair statistics, and they differ in the coherence they preserve: Section 7 turns that difference into a test. The quantum relaxation of the channels obtained by discarding the source record, the forgotten-source channels, is known exactly. Both laws have quantum spectral gap $(5-\sqrt{5})/60$ on the binary shell and on the full carrier, their slowest binary coherences decay by the sharp factors 5/6 per attempt for C and 7/10 for E, the binary-shell channel becomes entanglement breaking after finitely many attempts (134 attempts suffice by an explicit certificate), and on the full carrier it never does: an explicit partial transpose stays negative at every attempt count [67, 68].

2.4 What runs

The specification is executable, and its components run with retained receipts [67]. The companion construction paper states each component, its declared services and its certificates [66]; the following summary gives the parts used in this paper.

An observer that remembers and acts. The native observer executes two repair events with a finite tape. It records the first current, issues a forecast from that record, chooses the next seam from the record alone, records the result and halts. Exhaustive checks retain all 264 binary-source paths, and a 20,000-trial seeded execution reproduces the exact predictions. Record-controlled feedback gives final port-zero occupation 35/264, against 1/6 when the selector is disconnected or shuffled with the same action frequencies, so the response is caused by the retained record and not by the action statistics. Feedback preserves the committed classical record while it can reduce coherence between different record values. Complete records of these short histories recover the source optimally, with entanglement fidelity $D^{-2} \sum_h w_h \operatorname{rank}(P_h)^2$ for the maximally mixed input, where D is the number of carrier configurations, h runs over the complete admitted histories, w_h is the product of their seam-selection and coin probabilities and P_h projects onto the inputs compatible with h; it equals 1301/2028 for the full-carrier two-event C observer and 3329/6084 for E under the same current-sign selector [69].

An electromagnetic detector driven by the closure coordinate. A second observer evaluates the coupling $\alpha(P)$ from the screen grain P and sets $g(P) = \sqrt{4\pi\alpha(P)}$. A prepared path qubit reads the first current through a calibrated loop exposure χ; the detector bit $y = \pm 1$ has probability $[1 + y \sin(g(P) j_1 \chi)]/2$ and alone chooses the second seam, giving mean transferred charge $\mathbb{E}[g(P) j_2] = (5g(P)/264) \sin(g(P)\chi)$. All 528 weighted histories are retained and a 200,000-trial execution agrees with the analytic response. The closure coordinate thereby enters an actual response magnitude; the charge assignment, probe, exposure and controller are declared apparatus.

Fields measured by the observer. A scalar detector prepares a signed field intervention from the first native current, reads the field with a binary Weyl instrument and feeds the bit back, with

inherited continuum bounds transferred to the complete history law. A source-compiled two-site field apparatus builds its entangled preparation from blank registers, evolves the field and reads it with a quantum pointer, all from emitted twelve-port control words; at the executed sixteen-level cutoff an exact rational interval places the ideal plus probability in $[0.51495, 0.51789]$, excluding a fair detector.

One interacting process. The strongest joined execution puts quantum electric flux, a charged fermion pair and two neutral scalar sites on one work register. Native source words implement their noncommuting evolution, the matter and scalar pointer instruments, and feedback controlled by a committed matter bit; the final scalar bit controls the next native repair, and all 1,056 native and quantum histories retain their terminal states. For zero native preparation current the scalar feedback contrast of the same declared action has the certified infinite-occupation enclosure $[-0.024045670143, -0.024043450955]$, including the original free vacuum, unbounded interacting evolution and the actual coordinate half-line detector [70]. Section 4.3 develops what this process says about measurement.

One history under every schedule. The same sharp-readout history, written as local compatibility constraints, is reached under every fair update schedule, so observer states and modular readouts are functions of the completed history and the solver counter is auxiliary. The rank-three carrier of Theorem 4.1 attaches to all joint histories as a displacement readout, and fixed seam schedules propagate influence at most one graph edge per causal layer.

These executions are finite and their apparatus, services, field action and preparations are declared. They discharge the executable-observer row of the acceptance criteria under their declared services and inputs; longer observers with memory, reset and continuation contracts, and the continuum and gravitational rows, are not constructed (Section 5).

3 What observers agree on

Two observers who share a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records carry the familiar quantum rules.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\mathrm{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 6. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [4, 6].

3.1 Observable normal forms

Definition 3.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 3.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [4].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q + 1$ attempts, a bounded-waste upper horizon of $(q + 1)$ times the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n + 1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [4, 6].

3.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 3.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 3.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 3.3, so that it has the one-step diamond property, strictly decreases the quadratic descent functional V of Section 2, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 3.3 makes it locally confluent, so Newman's lemma [41] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [2]. The

level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. Across the rungs from five to nine, one initial load per rung gives an excess-energy contraction between 0.81 and 0.83 per sweep, and the sweep counts to the balanced class grow slowly with the carrier count, from 51 to 79 at level six to 84 to 100 at level ten; these finite readings prove no all-size mixing or termination-time bound. On the sixteen level-six schedules the number of unit transfers to the balanced class lies between 924,832 and 925,708, about 1.25 times half the initial ℓ_1 deviation of the loads from their mean. The first-settling configuration retains spatial information that its public component multiset does not encode: at level ten the initial-to-terminal amplitude responses of equal-count ancestral groups of sizes 1,024, 4,096 and 65,536 are 0.90758, 0.95232 and 0.98983, with conditional schedule-noise fractions 1.241, 0.594 and 0.143 percent. Stationary continuation adds a constructive fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [7].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations, and tetrahedral and octahedral controls recover their distinct specifications [2]. Complete federation gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

3.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional $*$-algebra with a normalized state. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public.

Theorem 3.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\, \rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{3}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0+B_1) + A_1(B_0-B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [22, 23]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1,1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [24], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [5]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [20, 21]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map [27, 5]. The same surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [25, 26]. Finite webs do not by themselves supply the valuation hypothesis: a source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity is the missing step [5].

4 Selected results

Five results are developed here, three as theorems with proofs and two as certified computations with their declared inputs. They were chosen because the executable construction consumes them or because they carry the programme's most specific structural content. Section 4.7 lists the other results with their classes.

4.1 Three spatial directions

The carrier incidence fixes the dimension of the carrier's own position readback, which is the reason the number three occurs in this framework without being declared.

Theorem 4.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1 - (5 - \sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (1) carried by the port position vectors. The same number $(5 - \sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 2.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint

histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [67]. Reading the completion as physical position consumes the realization map (M1) of Section 5.2 [1, 7].

4.2 The gauge algebra and one generation

Let A be the icosahedral adjacency operator and J the antipodal involution on the twelve ports. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \tag{4}$$

and the real port module splits as $P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'$: the even coordinates carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates the rank-three frame channel and its Galois-conjugate companion.

The theorem below uses two premises stated in the introduction and developed in Section 6.2. The carrier layer of the specification supplies a complete reversible response: an injective response derivative $D : P_{12} \to \mathfrak{u}(H)$ whose image is commutator-closed with a positive trace pairing. The no-outside rule applied to transformations requires every proper recharting of the carrier to be implemented from inside that response; the library calls this the endogeneity clause of A2.

Theorem 4.2 (Gauge Lie-type forcing)**.** *Let $D : P_{12} \to \mathfrak{u}(H)$ be the complete reversible response derivative of the carrier, and put $\mathfrak{g} = D(P_{12})$; let G_D^0 be the connected subgroup of the unitary group of H generated by $\exp \mathfrak{g}$. Suppose every proper carrier action $a \in A_5$ has a projective implementer satisfying*

$$\mathrm{Ad}_{[U_a]}\, D(v) = D(a \cdot v), \qquad [U_a] = [g_a c_a],$$

where $g_a \in G_D^0$ and c_a centralizes $\mathfrak{g}$ pointwise. Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The center is the unique trivial line. Up to exchanging $\mathbf{3}$ and $\mathbf{3}'$ by the outer automorphism of A_5, the $\mathfrak{su}(2)$ ideal carries $\mathbf{3}$ and the $\mathfrak{su}(3)$ ideal carries $\mathbf{3}' \oplus \mathbf{5}$.

Proof. Injectivity of D gives $\dim \mathfrak{g} = 12$, and the positive trace pairing makes the commutator-closed image a compact Lie algebra. For every $a \in A_5$ the centralizer factor acts trivially on $\mathfrak{g}$, so the action is inner, and covariance through the injective map D gives a homomorphism from A_5 to $\mathrm{Int}(\mathfrak{g})$. The transitive proper-carrier action fixes only the uniform port line, hence $\dim \mathfrak{g}^{A_5} = 1$. Compactness gives $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$, and inner automorphisms fix the center pointwise, so $\dim \mathfrak{z} \le 1$. If $\mathfrak{z} = 0$, the classification of compact simple Lie algebras forces $\mathfrak{su}(2)^4$ [58]; inner automorphisms preserve each ideal, and on each three-dimensional ideal A_5 acts trivially or faithfully, so its fixed space has dimension three or zero, since a finite rotation group fixing an axis is cyclic. The fixed-space dimension of $\mathfrak{g}$ would be a multiple of three, a contradiction. Hence $\dim \mathfrak{z} = 1$, the semisimple part has dimension eleven, and the classification leaves only $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. The three-dimensional ideal cannot carry the trivial action, so it carries $\mathbf{3}$ or $\mathbf{3}'$, and the eight-dimensional ideal carries the complementary triplet together with $\mathbf{5}$. □

The argument has a short intuitive form. Twelve ports give twelve independent ways to push the carrier, and complete response turns them into a twelve-dimensional compact Lie algebra. Internal implementation makes every recharting act by an inner automorphism, and inner automorphisms leave the center pointwise fixed, so the center lies in the directions the carrier symmetry fixes. Only the uniform push is fixed. The center is therefore at most one-dimensional, the remaining eleven dimensions must be semisimple, and $11 = 8 + 3$ is the only compact split.

No ambient compact group or candidate Lie algebra occurs among the premises. The port count twelve equals the dimension of the forced algebra, so the carrier fixes that dimension by declaration; the content of the theorem is the exclusion of the only other twelve-dimensional compact type $\mathfrak{su}(2)^4$ through compactness and the one-dimensional fixed space, together with the forcing of the unique center. The reconstruction of D from ordered reversible histories is consumed as a premise, and laboratory currents consume the realization map (M3) of Section 5.2.

The forced algebra has an explicit finite realization. Choose from each antipodal axis one of the certificate's exact coordinate vectors $u_i \in \mathbb{Q}(\sqrt{5})^3$, each with $\|u_i\|^2 = 2+\varphi$ for $\varphi = (1+\sqrt{5})/2$, and put $U = [u_1 \ \cdots \ u_6]$. For a real port field f with antipodal involution J, write

$$b_i = \frac{f_i + f_{Ji}}{2}, \qquad d_i = \frac{f_i - f_{Ji}}{2}, \qquad c = \frac{1}{6}\sum_i b_i, \qquad b^0 = b - c\mathbf{1},$$

so that the even coordinates b carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates d carry the rank-three frame channel detected by U and its Galois-conjugate companion detected by $\sigma(U)$, where σ is the field automorphism $\sqrt{5} \mapsto -\sqrt{5}$ applied entrywise. Let

$$\Phi_0(b^0) = \sum_{i=1}^{6} b_i^0 u_i u_i^{\mathsf{T}}, \qquad \widehat{x}\, y = x \times y.$$

The conditional matrix witness uses the space

$$H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W,$$

two copies of $\mathbb{C}^3$ carrying the exact rotations R_g and their Galois conjugates $\sigma(R_g)$ defined below, and four nonzero rational coefficients $\lambda_1, \lambda_5, \lambda_3, \lambda_{3'}$, one on each source band. These are explicit branch premises. The two three-dimensional response blocks match the two rank-three odd channels; they are not asserted to be a derived physical Hilbert space. Write v_p for the signed coordinate vector at port p. For $g \in A_5$, let $R_g \in \mathrm{SO}(3)$ be the exact rotation satisfying $R_g v_p = v_{g(p)}$ on the oriented port frame, and define

$$\Pi(g) = \mathrm{diag}(R_g, \sigma(R_g)).$$

The target A_5 action on $\mathfrak{u}(H)$ is conjugation by $\Pi(g)$.

Theorem 4.3 (Conditional four-band finite port-response algebra)**.** *On the declared twelve-port carrier and supplied response representation, define*

$$\begin{aligned} K(f) = \mathrm{diag}\big(\lambda_3 \widehat{Ud} + i\{\lambda_1 c I_3 + \lambda_5 \Phi_0(b^0)\}, \\ \lambda_{3'} \widehat{\sigma(U)d}\big) \in \mathfrak{u}(H). \end{aligned} \tag{5}$$

Then K is injective, its image is commutator-closed, and

$$K(g\cdot f) = \Pi(g) K(f) \Pi(g)^*.$$

The induced A_5 action on $\operatorname{im} K$ is inner. Let $K^{-1} : \operatorname{im} K \to P_{12}$ denote the inverse of K onto its image. With the bracket pulled back from that image,

$$[x, y]_K = K^{-1}\big([K(x), K(y)]_{\mathrm{mat}}\big),$$

one has

$$(P_{12}, [\,\cdot\,,\cdot\,]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{6}$$

Its center is the uniform one-dimensional port line, its derived algebra has dimension eleven, and the five-dimensional A_5 band is noncentral. The map has a positive-definite invariant Hilbert–Schmidt pullback.

Proof. Evaluating (4) on the four adjacency eigenspaces gives the antipode relation, and the source protocol solves the common farthest-shell filter to give the signed response $-J$. The constant coordinate maps to $i\mathbb{R}I_3$; the six rank-one axis matrices span the symmetric 3×3 matrices with $\sum_i u_i u_i^{\mathsf{T}} = (5+\sqrt{5})I_3$, so their sum-zero coefficients map isomorphically onto the traceless symmetric matrices under Φ_0; U is an isomorphism on the **3** band and annihilates $\mathbf{3}'$, with $\sigma(U)$ complementary. Hence K has rank 12 and image exactly $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Equivariance, positive definiteness of the invariant pairing, and the statement that $\Pi(g)$ is the exponential of an element of $\operatorname{im} K$ are exact calculations in $\mathbb{Q}(\sqrt{5})$ over the full sixty-element domain, part of the proof by the exhaustive-computation rule of Table 1 [1]. □

Boundary 4.4. The A_5-module decomposition alone does not determine a Lie bracket: the same module also carries the zero bracket, whose trivial response-generated group fails the A2 endogeneity clause. The independent inputs to Theorem 4.3 are the carrier incidence and orientation, the supplied response representation, and four nonzero band coefficients. The block architecture and coefficients are declared. No measured particle, mass, or coupling value enters the exact verification. The theorem constructs the pulled-back bracket for this response map; it does not select K from source histories or prove a uniqueness claim over all A_5-equivariant brackets or finite carriers. The abstract Lie type is forced independently by Theorem 4.2.

Theorem 4.5 (Finite matter image and exhaustive exterior-menu selection)**.** *Let $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$ be an explicitly supplied trace-balanced matter representation of the forced Lie algebra, with $\widetilde{G} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acting by $(g_3, g_2, z) \cdot (v_3, v_2) = (z^{-2} g_3 v_3, z^3 g_2 v_2)$. Its exterior module branches as*

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \tag{7}$$

with $Q = (3,2)_{1/6}$, $u^c = (\bar{3},1)_{-2/3}$, $e^c = (1,1)_1$, $d^c = (\bar{3},1)_{1/3}$ and $L = (1,2)_{-1/2}$: fifteen states, all perturbative gauge and mixed gravitational anomalies vanishing, and an even number of weak doublets. The common kernel of the action is $\mathbb{Z}_6$, so the maximal faithful image is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{8}$$

*Among all $2^{10} = 1{,}024$ subsets of the ten nontrivial irreducible summands of $\Lambda^\bullet V$, imposing nonemptiness, chirality and vanishing of the mixed gravitational–*U(1)*, $\mathrm{SU}(3)^2\mathrm{U}(1)$, $\mathrm{SU}(2)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ traces leaves exactly two rank-fifteen masks, exchanged by charge conjugation.*

Proof. The exterior-sum identity gives the five components and charges, and the anomaly sums are the standard ones [62]: for example $6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 1 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 = 0$, and with three color copies of Q against one L the Witten parity is even [64]. The element $(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3})$ acts trivially and generates a cyclic group of order six, and center-action enumeration excludes a larger kernel. The scan evaluates the four traces on all 1,024 masks; its only survivors are the two exterior-parity masks. □

The branching is the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ of SU(5) [61], and the quotient is a possible global form of the Standard Model group [63]. One corollary concerns a channel the framework does not contain: the forced adjoint $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$ contains no mixed $(3,2,-5/6)$ generator, so the minimal simple grand-unified exchange channel is absent from the twelve-dimensional current algebra. Separately, Doplicher–Roberts reconstruction supplies a compact sector group from realized bosonic edge sectors [59, 60]; its equality with the finite response group is a separate commuting-square premise. Physical matter identification, family multiplicity, Yukawa couplings and masses consume (M3) and (M5) [11].

4.3 Measurement inside one interacting process

OPH places measurement at the transition from a private repairable record to a protected public normal form: a record is public when it has survived every authorized comparison and no repair can change it, and event projectors, weights and updates act on the completed algebra-state surface of Section 3.3. The executable observer puts this account inside one interacting quantum process [66].

Quantum electric flux, a charged fermion pair obeying the canonical anticommutation relations, and two neutral scalar sites share one work register. Native source words implement their non-commuting evolution with the Gauss law holding as an operator identity, a matter pointer writes a committed bit, feedback controlled by that bit acts on the field, and a scalar pointer writes a second committed bit that controls the next native repair. All 1,056 native and quantum histories retain their terminal states. At scalar dimension two and 512 temporal steps per interval an exact certificate places the ideal feedback contrast in $[-0.033297300667839, -0.030025660061775]$, and for zero native preparation current the infinite-occupation contrast of the same declared action lies in $[-0.024045670143, -0.024043450955]$ [70]. The detector of the second interval is the actual coordinate half-line projector; taking the sign of a finite coordinate matrix would be a different instrument.

Accurate outcome probabilities do not certify an accurate post-measurement state. For the same zero-current preparation, the complete retained output of the sharp detector at occupation dimension two lies at trace distance about 0.455 from the infinite sharp history, although its sign probability moves far less, and every output confined to the dimension-256 occupation code has error above 0.122. For the vacuum control the optimal error within these occupation codes decays only as $d^{-1/4}$, so an output error ϵ requires dimension of order ϵ^{-4} [71]. A finite-resolution Ramsey detector on the same action has a quantitative error bound for its complete history, including the measurement, and an infinite-model measurement energy increment of exactly $1/8$ in the declared units.

The same process supplies the observer's own descriptions. The GNS representation of the actual observation state yields nontrivial modular flow on a declared observer algebra, and a source-executed binary clock probe reads that flow while retaining every earlier record. A faithful Gibbs reference for the full Hamiltonian identifies continuous between-event evolution with modular flow by $t = -\beta s$, while the actual observed state changes. Written as local compatibility constraints, the completed history is reached under every fair update schedule, so observer readouts are functions of the completed history. How much of its past a record keeps is exact as well: on the binary shell the forgotten-source channel eventually breaks entanglement, on the full carrier it never does, and complete native records recover the source with the optimal fidelity of Section 2.4 [68, 69].

The field action, preparation, coupling and model-time schedule are declared construction inputs, and the results concern this two-site action; a spatially refining family is the connection that Section 5 lists as not constructed.

4.4 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\mathrm{Conf}^{+}(S^2) \;\cong\; \mathrm{PSL}(2,\mathbb{C}) \;\cong\; \mathrm{SO}^{+}(3,1), \qquad H^3 \;\cong\; \mathrm{SO}^{+}(3,1)/\mathrm{SO}(3), \tag{9}$$

which identify the three-dimensional space of future unit timelike directions [47]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on,

and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [37]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1].

Theorem 4.6 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t, x) = t^2 - g_{\mathrm{src}}(x, x)$.*

(F1) W_{src} has dimension $1 + 3 = 4$ and Q has one positive and three negative directions;

(F2) source-unit directions, S^2 and future-null rays are exactly equivalent;

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ sends every generated precedence into the future cone whenever each parent edge obeys $g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 (h_{\mathcal{L}}(f) - h_{\mathcal{L}}(e))^2$;

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ is spacelike, then $e \preceq_{\mathcal{L}} f$ holds exactly when $F(f) - F(e)$ is future causal, so F is an exact order embedding preserving every causal interval.

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n + 1$, n the number of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 9]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [28, 29, 30].

Proposition 4.7 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [35, 36]; the local model is $(1 + 3)$, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [33, 34], an explicit*

tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with G_{ab}, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$.

A source-record population realizes the flat case.

Theorem 4.8 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let $\varphi = (1+\sqrt{5})/2$, $L = 2/\sqrt{\varphi+2}$ and $q = F_n$ the n-th Fibonacci number, $n \geq 4$. The population $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0,L]^3$ consists of q^3 distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within $a_q = L/\sqrt{q}$, with model duration $\Delta_q = a_q/c$ for a declared speed c. As $q \to \infty$ the order converges away from null separation to the speed-c causal order, and with $\rho_q = q^3/(L^3\Delta_q)$, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q counts strict causally ordered pairs; for vertical duration T, $\mathrm{Vol}(I) = \pi c^3 T^4/24$.

Corollary 4.9 (Proper-time ratios from record counts)**.** *If I_q, J_q approach interior timelike diamonds of proper durations $\tau_I, \tau_J > 0$, the observer readout $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at $q = 13, 21, 34, 55, 89$ [31, 32]; the outermost central diamonds, which the cube clips and which are therefore not interior readings, read 4.15, 3.61, 4.10, 4.09 and 4.07, and the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [9].

4.5 The Einstein equation from a realized null balance

Theorem 4.10 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 4.4; nothing below depends on the choice. If a symmetric form X satisfies $X(k,k) = 0$ for every null vector k, then $X = \phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1, \pm e_i)$ and $(1, s, s, \pm s)$, $(1, s, -s, s)$ with $s = \sqrt{3}/3$, reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ with design determinant $8192/27$ in the certificate's basis and a decoder obeying $\|x\|_\infty \leq (2+\sqrt{3})\|q\|_\infty$.*

Proof. On future null rays $(1, n)$ the identity $X_{00} + 2X_{0i}n_i + X_{ij}n_in_j = 0$ on the unit sphere forces $X_{0i} = 0$, $X_{ij} = \phi\delta_{ij}$ and $X_{00} = -\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 9]. □

Theorem 4.11 (Tensor completion of a realized null balance)**.** *Let (M, g) be a connected, time-oriented, C^3 Lorentzian four-manifold and T_{ab} a symmetric C^1 tensor with $\nabla^a T_{ab} = 0$. Suppose a common realized tower supplies $\kappa = 8\pi G_N > 0$ with*

$$G^{(g)}_{ab} k^a k^b = \kappa T_{ab} k^a k^b \tag{10}$$

for every null vector k at every point. Then there is a constant Λ with

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}, \tag{11}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ fixes $\Lambda = \Lambda_0$.

Proof. Theorem 4.10 applied pointwise to $G^{(g)} - \kappa T$ gives $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b \phi = 0$, connectedness makes ϕ constant, and $\Lambda = -\phi$. □

The structure of the argument is simple. Light rays cannot see a stress term proportional to the metric, so balance between curvature and stress on every null direction fixes Einstein's equation up to such a term, and the Bianchi identity with stress conservation makes its coefficient constant. That constant is the cosmological constant. The coefficients of entanglement-equilibrium arguments [54] enter through one machine-checked identity: scalars with $\delta S_{\text{bulk}} = \frac{8\pi^2 \ell^4}{15} t$, $\delta A = -\frac{4\pi \ell^4}{15} f$ and $\delta S_{\text{bulk}} + \delta A/(4G_N) = 0$ obey $f = 8\pi G_N t$. For OPH to supply (10), one common refinement tower must realize (G1) the effective manifold of Proposition 4.7; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L \rangle = \delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa = 8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

4.6 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular charge, with anomalous modular energy on overlap collars as one candidate carrier [14]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit counterexample. A per-cut model gives $a_0 = Gn^2 c$, with only the product $n^2 c$ identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\text{DE}} = \kappa/N$ at constant κ gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [15].

4.7 The wider inventory

Table 2 lists the structures of observed physics that the reconstruction returns, with the class of each result and where it is proved or stated. A finite theorem is a theorem about the declared finite objects, a reconstruction implication carries the physical antecedents displayed with it, and a diagnostic consumes measured input and predicts nothing.

Table 2: Reproduced structures of observed physics, the result that returns each one, its claim class, and its location.

Structure	**What the reconstruction returns**	**Class**	**Where**
An observer that records and acts	A finite observer executes repair, retains its records and chooses later interactions from them; feedback changes the outcome law against disconnected and shuffled controls	certified finite computation	§2.4
Measurement inside an interacting field process	Flux, charged matter and a scalar field on one register with complete instruments and feedback; a certified nonzero infinite-occupation response	certified finite computation	§4.3
Schedule-independent public facts	Every maximal asynchronous repair schedule reaches the same protected normal form under the transactional local diamond	finite theorem	§3
Probability weights on outcomes	Every nonnegative normalized valuation additive on coexisting effects is the trace against a unique density operator, in every finite dimension, with no continuity axiom	finite theorem	§3.3
State update after an outcome	A swap-twisted instrument has the same effects, so the effect table does not select the update; invariance on the outcome-certain states selects the Lüders map	finite theorem	§3.3
The correlation ceiling $2\sqrt{2}$	Tsirelson bound on commuting record subalgebras, attained exactly on a declared entangled witness, with record-diagonal states confined to 2	finite theorem	§3.3
Impossibility of copying an unknown state	One ancilla-free isometry copying two sharp states from a common blank forces their overlap to 0 or 1	finite theorem	§3.3
A three-dimensional position carrier	The normalized twelve-port repair response has an intrinsic rank-three Gram quotient into which the record modules embed densely	finite theorem	§4.1
Lorentz kinematics and the velocity space	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ with H^3 the space of future unit timelike directions	classical identification on the supplied spherical support	§4.4
A source-derived $1+3$ causal precursor	The real axis and the rank-three source quotient give Lorentz inertia $(1,3)$ with the S^2 null boundary, and authenticated logs embed by exact order embeddings	finite theorem plus reconstruction implication	§4.4
Proper time from counting	Record counts in source-record diamonds recover proper-time ratios by the unique fourth-root readout	finite theorem	§4.4

Table 2 continued

Structure	What the reconstruction returns	Class	Where
The Einstein field equation with Λ	Tensor completion of a realized null balance under Ward conservation, the Bianchi identity, and one scale identification	reconstruction implication	§4.5
The Standard Model gauge Lie type	Complete response with internally implemented rechartings forces $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ on twelve ports	finite theorem	§4.2
One anomaly-free generation of fifteen states	Exhaustive scan of the declared exterior menu leaves one charge-conjugate pair of rank-fifteen chiral anomaly-free masks, with common kernel $\mathbb{Z}_6$	finite theorem	§4.2
Absence of the minimal grand-unified exchange channel	The forced adjoint contains no X/Y generator	finite theorem	§4.2
The relativistic mass shell and internal energy in inertia	$E^2 = m^2 + \lvert p\rvert^2$ for the declared-mass momentum, and two declared selection routes giving inertial coefficient $m + E$	finite theorem	[8]
A nonlinear matter continuum and an interacting state space	First-order trajectory convergence for the real quartic-wave sector of a supplied charged action, and a unique self-adjoint closure of its declared Hamiltonian	conditional analytic theorem	[8]
The arrow of time and the four laws	Reference-preserving stochastic repair contracts relative entropy; one conditional package gives the four laws, fluctuation relations and the Landauer bound	finite theorem	[8]
Entropy that scales with a boundary	Collar decomposition with one-sided edge term $\sum_\alpha p_\alpha \log d_\alpha$; the sparse crossing-read functional Γ-converges to perimeter	finite theorem; conditional analytic theorem	[9]
Coulomb and Maxwell structure	Unique minimal-energy neutral Coulomb solution, a local twenty-face curvature action gauge invariant for conserved seam currents, and a hopping charge conserved at a declared nonzero step h	finite theorem	[7]
Massless photon, gluon and graviton	Three conditional structural statements with zero hard mass parameters on their declared branches	reconstruction implication	[11]
A positive Yang–Mills mass gap	The continuum gap identified with the limiting repair gap under named finite-rate, transfer and reconstruction certificates	reconstruction implication	[16]
The Koide relation	Positive-chamber circulant identity with $Q = 2/3$ exactly at $\rho/a = 1/\sqrt{2}$, with a custody-bound tau window	finite theorem; prospectively fixed conditional test	[12]
The fine-structure constant	Interval-certified screen-grain roots 136.994835 and 137.035660 against 137.035999	diagnostic	§6.5
The cosmological constant	Capacity candidates 3.2921×10^{122} and 3.3001×10^{122} against 3.3129×10^{122}	diagnostic	§6.5
Flat rotation curves and the baryonic Tully–Fisher relation	Under deep scale covariance and quadrature composition the unique law gives $v^4 = GM_b a_0$; a_0 is not source-derived	conditional characterization theorem	§4.6

Table 2 continued

Structure	What the reconstruction returns	Class	Where
The dark-energy equation of state	Fixed record capacity under $\rho_{\mathrm{DE}} = \kappa/N$ gives $(w_0, w_a) = (-1, 0)$	reconstruction implication	§4.6
The de Sitter transfer sign	Exact capacity identity with $\Delta S_{\mathrm{gen}}^{\max} = \log(1-f) < 0$	finite theorem	[13]
Isotropy of propagation to angular rank five	The first symmetry-allowed anisotropic rank of the carrier class is six, with two frozen rays inside one exact band	finite theorem; frozen prospective branch prediction	§7.1
An instrument comparison fixed before data	C and E repair predict 5/8, 1/8 and 3/8 on fixed coherent settings, with a complete-count rejection rule	prospectively fixed conditional test	§7

5 The specification and its scoreboard

5.1 The completion target

The programme has one target: an executable realization of the observer algebra whose compatible inter-observer measurements reproduce the observed universe. Observers inside the construction must make the measurements we make and must be able to use the effective descriptions we use: their joint statistics must admit a nontrivial $3+1$-dimensional quantum field theory as a controlled effective description, including the observed matter and interactions, and classical physics and dynamical gravity under appropriate coarse-graining. No fundamental field or spacetime has to be reconstructed first. Fields, particles and geometry can be effective descriptions of what observers measure and jointly interpret, valid in their regimes, without being primitive objects. Exact equality of limiting observable laws is the mathematical target; finite executions and experiments are judged with declared approximation and measurement errors.

Table 3 gives the acceptance criteria and the status of each. "Discharged" means executed with retained receipts or proved on the stated domain; "supplied" means consumed as a declared input of the construction that uses it.

Table 3: Acceptance criteria of the completion target and their status.

Obligation	Acceptance evidence	Discharged	Supplied or not constructed
Executable observer algebra	Explicit state, operation, preparation, measurement, retained-record and feedback maps, executed, with accounted memory and services	Native observer, electromagnetic and scalar detectors, source-compiled field apparatus, the common interacting process and both C/E instruments, each with retained receipts	Initial ensembles, apparatus and services are declared; longer observers with memory, reset and continuation contracts are not constructed

Table 3 continued

Obligation	Acceptance evidence	Discharged	Supplied or not constructed
Shared operational meaning	Interpretation maps fixed independently of validation outcomes, compatible across observers and refinement, preserving intervention statistics and quantum correlations	Schedule-independent public records (Theorem 3.4); the completed interacting history under every fair schedule	Interpretation maps for physical quantities across observers and refinement are not constructed
Effective quantum-field description	A refining family of observer statistics whose errors relative to a nontrivial $3+1$-dimensional QFT tend to zero, including observed matter and interactions	One interacting register with operator Gauss law; infinite-occupation and complete-output certificates for a two-site action; free-field continuum theorems	The field action, vacuum and coupling are supplied; a spatially refining interacting family is not constructed
Classical and gravitational descriptions	Compatible classical, geometric, stress and clock descriptions of the same observations, reproducing gravitational predictions with controlled corrections	Rank-three displacement readouts, causal cones, the finite $(1+3)$ precursor, flat count volume, tensor completion of null balance	Read law, model clock, stress, entropy and scale premises are supplied; responsive geometry on the same process is not constructed
Physical predictions	Input ancestry, calibration and decision rules fixed before independent data, agreement with observed statistics	The C/E protocol and the angular-rank-six band are fixed before comparison	Physical encoding, service calibration and independent data are not supplied

5.2 The standing realization maps

A statement about nature needs one physical realization carrying compatible events, fields, action, states and readouts across refinement. Five maps name what such a realization must supply, and each boundary statement in this paper cites them by label.

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit.

(M2) Clock and energy calibration. An identification of the internal step with a laboratory clock and of the modular ledger with a laboratory energy, in measured units; configuration-derived model duration does not select laboratory time or its units.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) A source-selected action. An action produced by the source rather than declared, with its field sector, coupled interaction, relative normalizations, and operational readout.

The executable observer turns these maps into construction work on one process. Its retained histories carry a rank-three displacement readout, modular clock probes and committed records on a single register; what such a realization lacks is a spatially refining family of such processes whose records, clocks, currents and stress readouts converge together. A finite theorem that consumes none of the five maps stands on its own domain, and no result in this paper discharges any of them.

5.3 Open connections

These connections can be built in parallel, and each concerns the same process:

(i) connect a spatially refining family of source programs to the continuum while retaining their records and bounding numerical and physical service errors;

(ii) attach interacting matter, responsive geometry and physical clock calibration to the same state, event law and retained records;

(iii) implement longer and composed observers with explicit memory, reset, continuation and resource contracts;

(iv) realize the fixed C/E preparation and observable, establish its service budget independently and collect complete records for the declared rejection protocol.

The construction paper gives the detailed premise map, including the retained negative controls that bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached [66].

6 Closure: a universe that reads itself into existence

The previous sections built observers, their agreement and the structures they can read off, and they stated the executable specification as a candidate with a scoreboard. This section returns to the proposal stated in the introduction and develops it in full. It explains why the specification is exempt from a-priori justification and why the theory has no first premise, although its exposition has an entry point.

6.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}, \tag{12}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered [72]. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

No external computer runs the history and no additional time exists in which the whole is played. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration. Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x+1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x+2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, uniqueness and the identification of our universe with a fixed point are questions the equation leaves to the programme. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms [6]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [8].

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence [72]. A playback can match every frame of one history and fail this test as soon as a retained state is changed. The controls of Section 2.4 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

6.2 No outside

A theory of an external world must justify its postulates by an argument given in advance or by fit to data gathered from outside the theory. A self-contained structure offers no vantage point from which a premise could be imposed. This changes what a premise has to do, and it gives one rule with three applications.

Justification. A premise is justified when observers recover it and the specification that implements it runs and reproduces their measurements. The carrier, the observer algebra, the repair law and the closure coordinates are found by observers inside the structure and built by them; asking for an additional a-priori reason would ask for a view from outside, which the hypothesis denies. Construction choices therefore need no uniqueness theorem. What they owe is execution and agreement with what observers measure, which is the content of Section 5.

State selection. The rule also forbids assuming what observers have not recovered. After every observer-visible constraint has been imposed, the selected state may carry no further information relative to the recovered reference, so it minimizes the weighted relative entropy $\sum_p w_p D(\rho_p \| \tau_p)$ over the compatible local state families of the patches p. The theorem library calls this premise A3. Its finite objective has exact consequences: every minimizing local density has the union of the feasible supports, and a feasible witness with failure probability $b > 0$ gives an explicit floor on the selected probability [9].

Transformations. Every change of description that observers can make must be implemented by the structure's own operations, since nothing else could implement it. On a carrier this means

that each proper recharting is realized from inside the complete reversible response. The library calls this the endogeneity clause of A2, and it carries real weight: Theorem 4.2 uses it to exclude the only other twelve-dimensional compact type and to force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$.

6.3 Agreement and the meaning of "our"

The rule says what needs no outside justification. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 3.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: the shared-meaning row of Table 3 requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. The theorem library calls the interpretation clause of this condition A2. The basis reference states the hypothesis, the rule, the condition and the specification formally, with the library's premise statements and their dictionary [73].

6.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. The order of justification runs the other way: the hypothesis comes first, since it is what exempts the specification from outside justification. The structure itself has no first element in either sense. A pair of mutually determining equations such as $x = (y+2)/2$, $y = (x+2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two [72].

No temporal paradox arises. The closure equation relates parts of one timeless object, with no global time coordinate along which information travels backward. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. Repair descent is a well-founded order on authorized updates used to prove termination and compare schedules; it is neither an observer's experienced time nor a universe-level clock.

What is found and what is made become two roles in one structure. A rule invented for a game has consequences its inventor discovers. Here the architecture that observers discover constrains what they can build, and what they build belongs to the structure from which the discovery is made. The same reading answers the question why anything exists at the level the hypothesis can answer it: the question includes its questioner, and a complete inhabited structure whose dependencies constitute its operation needs no further act of actualization [72]. A demand for something in addition to the complete relational structure is, on this reading, misplaced. If several closures qualify, the hypothesis admits several worlds, and self-consistency alone has not selected ours.

6.5 What closure fixes, and what observation reads off

The architecture carries two dimensionless closure coordinates: the screen grain P and the record capacity N. Under the closure hypothesis these coordinates are fixed points of the closure map, and the loop also licenses reading them off: observers locate a constant by measurement, declare a map, and certify a fixed point of that map inside the measured basin. Whether a certified fixed point coincides with the measured value is a diagnostic in the sense of Table 1, not a consequence

of the hypothesis. The theory stands on its recovered specification whether or not a first-principles evaluation of every coordinate is found.

The screen-grain loop

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling a declared map returns at trial grain P and $\varphi = (1+\sqrt{5})/2$, has one interval-certified fixed point per declared map. The bare source map settles at $\alpha^{-1} = 136.994835177413$, a relative distance of 3.0×10^{-4} from the 2022 CODATA value 137.035999177(21) [65]; the self-consistent map, which recomputes the finite-screen unified gauge width inside the loop, settles at $P = 1.6309682414$ and $\alpha^{-1} = 137.035660136947$, a relative distance of 2.5×10^{-6}, the gauge width supplying 99.2 percent of the gap. The residual is attributed to same-scheme low-energy transport to the Thomson endpoint, which neither map computes and which this paper does not supply [10]. Reading the capacity as de Sitter horizon entropy in nats gives

$$N = \frac{3\pi}{\Lambda\,\ell_P^2}, \tag{13}$$

with $\ell_P^2 = \hbar G/c^3$, and the capacity branch gives $N = 3.2921\times10^{122}$ and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck base-ΛCDM chain; in the constant they read $\Lambda = 1.0959\times10^{-52}$ and $1.0933\times10^{-52}\,\mathrm{m}^{-2}$ against $3\Omega_\Lambda H_0^2/c^2 = 1.0891\times10^{-52}\,\mathrm{m}^{-2}$ at the Planck base-ΛCDM centrals, distances of +0.62 and +0.38 percent. Equation (13) is the dictionary that produces the comparison coordinate, so this restates the capacity diagnostic in the constant, and it consumes the measured $\hbar$, G and c through the calibration map (M2). Neither clears the source-only selection and same-quantity gates. Both comparisons are diagnostics of declared maps in the sense of Table 1: the maps were exposed before the comparison, the choice of map is itself declared, and neither comparison carries predictive weight. The executable detector of Section 2.4 consumes P through the coupling $g(P) = \sqrt{4\pi\alpha(P)}$, and N bounds the correctable records of a declared capacity carrier by $\log M_0 \le N$.

7 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [3]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer.

7.1 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [49].

Theorem 7.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1-\cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank*

five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order a^4k^6, suppressed by exactly $1/16$ under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 7.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 7.1 with scale-independent coefficients, the complete orbit of the twelve port directions u_i is its sole hop support through order k^6, the carrier scale a is finite and positive with the quadratic term normalized to k^2, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}\bigl[1-\cos(ak\,u_i\cdot\widehat{k})\bigr] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{14}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$, so in a transported frame the branch predicts the scale-free relations

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{15}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ vanishes.

Proof. The exact carrier moments $\sum_i(u_i\cdot n)^2 = 4$, $\sum_i(u_i\cdot n)^4 = 12/5$ and $\sum_i(u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 7.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module D_6 of Theorem 4.1, with Smith invariants $(1,1,1,1,1,2)$, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight $1/60$ on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Bigl[1-\cos\Bigl(ak\,w_j\cdot\widehat{k}\Bigr)\Bigr], \tag{16}$$

with $C_4 = -a^2/20$, $B_0 = a^4/840$ *and* $B_6 = -a^4/12600$, *hence* $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$ *and* $B_6/B_0 = -1/15$. *The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated* $\mathcal{I}_6$ *orbit.*

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 4.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j \cdot n)^2 = 10$, $\sum_j (w_j \cdot n)^4 = 6$ and $\sum_j (w_j \cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (16) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The declared symbol also carries an exact remainder certificate with no physical premise: with $q = ak$, $|\widehat{\Lambda} - P_6| \le \frac{7}{388800} q^8$ and $\frac{19}{20} q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600) q^6$; the certificate turns a into no SI length. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 7.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits* O_s *of the sixty proper carrier rotations with radii* $r_s > 0$, *weights* $w_s > 0$ *and the* k^2 *coefficient normalized to one, write* $\Lambda_{\mathrm{norm}} = k^2 + C_4 k^4 + (B_0 + B_6 \mathcal{I}_6(\widehat{k})) k^6 + O(k^8)$ *and* $\mu_m = \sum_s w_s |O_s| r_s^m$. *Then* $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ *for every member;* $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \ge 10/21$ *with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and* B_6/B_0 *lies in* $[-16/135, 16/75]$, *the face and vertex orbits attaining the endpoints and the edge orbit sitting at* $-1/15$.

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_i W_j r_i^2 r_j^2 (r_i^2 - r_j^2)^2 \ge 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21} C_4^2$, $B_6 = \frac{32}{315} C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [3].

7.2 An instrument test from the executable observer

The two native repair laws C and E have identical diagonal classical statistics and differ in the coherence they preserve. On a fixed coherent two-state source preparation, reversing the phase

changes the C readout probability from 5/8 to 1/8, while E predicts 3/8 for both. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of 1/64 and an exact sampling certificate below 1/1000; it rejects malformed or incomplete records and reports *no data* when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [66]. Synthetic controls exercise the decision rule and establish no physical verdict; physical encoding, independent calibration and comparison data are the open requirements.

7.3 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 6.5 are diagnostics of declared maps. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and algebraic results retain their stated domains under all three.

8 Machine verification and reproducibility

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations and outward-rounded interval certificates, each emitting a canonical hashed payload; executable observer receipts that replay every retained result and check its artifact hashes; simulation receipt bundles pinning the source revision, configuration, seed where stochastic, tolerances and output hashes; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees; and adversarial negative controls, so that every certificate ships with mutations that must be detected. Machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test or rejected candidate, and reject any stronger classification unsupported by the recorded dependencies.

Table 4 names the artifact family behind each group of results. The repository, receipts, schemas and rebuild instructions are public [1, 2, 67]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries; analytic arguments in this manuscript retain their displayed proofs, and no part of the stack establishes physical realization.

Table 4: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Executable observer, detectors, interacting process and C/E instruments	The observer receipts with exhaustive path replays, seeded executions, feedback controls, exact and interval certificates, and a single verification entry point that replays every receipt and checks its hashes [67, 66]

Table 4 continued

Results	Principal artifacts
Normal forms, stability, refinement, complexity and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born weights, Lüders selection, Tsirelson saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound and the public record algebra, with per-module axiom audits
Rank-three completion	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier
Gauge Lie type, matrix witness, matter image and exterior selection	The screen modules for the carrier action, commutant and incidence response; the exact-arithmetic port current certificate; the trace-balanced kernel and exterior scan modules
Causal precursor, count limit and proper-time readout	The provenance and geometry modules; the source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, and the public archive
Null tomography, small-ball arithmetic and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; explicit null-balance, Ward, Bianchi and scale premises
Angular-rank theorem, frozen branches and the dispersion band	The primitive-port prediction and carrier-class band modules with mutation tests; the custody-bound prediction register
Exact federation and the closure loop	The level-six configuration, seeds, seam arrays, loads, per-schedule ledgers, quotient hashes, standalone verifier and public archive; the level-ten settlement receipts; the event logs and recovered specifications of the canonical and control sources
Dark sector, vacuum coordinate and closure roots	The dark-sector, deep-profile, per-cut and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root

9 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [17, 18, 19]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion. Quantum Darwinism locates classical objectivity in redundant environmental records [44], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [45], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [46], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [42] and the thermal-time programme reads time from modular structure [43]; OPH attaches public facts to protected overlap records, and its executable observer reads modular flow from the actual observation state of a running process. The local-diamond proof sits beside asynchronous agreement [38, 39], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [40].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [53, 54, 57], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [50, 51, 52]. The route from a null-projected balance and the

Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [55, 56]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [59, 60] and contributes the forcing of the local Lie type from complete response and internally implemented rechartings.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [74]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [75], and Nozick proposed that a fundamental principle might subsume itself [76]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [77]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

10 Conclusion

OPH proposes that physics is what bounded observers can establish together, and that the universe is the structure in which such observers recover their own specification and realize it. Because nothing lies outside that structure, the specification owes no a-priori justification; it has to run and return the world it was recovered from. The specification runs. Its observer repairs, records and acts on its records, its detector draws its coupling from a closure coordinate, and one interacting quantum process carries flux, charged matter and a scalar field through complete measurements with a certified feedback response.

The finite theorems turn the specification's readouts into familiar structure: public records independent of the repair schedule, the Born rule and the Tsirelson bound on completed records, an intrinsic carrier with three directions, the Standard Model gauge Lie algebra with an anomaly-free generation, a source-derived $(1 + 3)$ causal precursor with proper-time ratios from record counts, and the Einstein equation from a realized null balance. The prospective content is the frozen dispersion band of Theorem 7.4 and Proposition 7.2: a resolved intrinsic dispersion with $C_4 \geq 0$, with $B_0/C_4^2 < 10/21$, with B_6/B_0 outside $[-16/135, 16/75]$ or with a nonzero coefficient at angular ranks one through five excludes every member of the declared carrier class at once; and the C/E instrument test, whose decision rule is fixed before any independent count.

The acceptance criteria state what is missing: a spatially refining family of interacting observer processes whose statistics converge to a $3 + 1$-dimensional quantum field theory with the observed matter, and responsive geometry with calibrated clocks on the same histories. The five maps of Section 5.2 are consumed rather than constructed, so these results do not establish that the architecture describes our universe. They specify what a realization must preserve, and the executable observer is the construction that has to carry it.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2048) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [4, 5, 6, 9, 11, 66], a relationship disclosed here so that overlap is visible.

Statements and Declarations

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Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the mathematical constructions, proofs, and counterexamples for the mechanism-variant comparison kernel and conditional example, protected-behavior first-hit obstruction profiles and finite-state transport, source-derived phase-effect selection, canonical adaptive-repair endpoint stratification, cumulative attempt-capacity bounds and sharpness results, and input-independent fixed computation-federation progress and execution. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav

Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

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Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 8.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [85, 87, 83, 84].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [79, 81, 88, 86, 82, 83, 84].
- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [80, 86, 83, 78].

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